F16-GE-001
ESSENTIAL SUPPLIES
11kV GENERATOR
2000-EL019-D04-10005

2.5MVA 11kV

F16GE001-NR
200/1A 2.5VA 5P20
30/0.05A
254Ω +/-5% 25A FOR 10s
SENSING ER-15kV

RTD

200/1A 2.5VA 5P20

PR ComAp NOTE 6

F16-SR-001
11kV FLOTATION SWITCHROOM

NOTE 8 NOTE 4

ACTIVATION OF SWITCHROOM E-SHUT (NOTE 5)

F16-SW-011
FLOTATION 11kV SWITCHBOARD

EARTH FAULT

TEST

200/1A 2.5VA 5P20

50 51 NOTE 2 AFD PR SEL 751

PR SE-330AU 50G NOTE 6A

200/1A 2.5VA 5P20

CLOSE TRIP FAULT TRIP

630A

27 59 46

PR SEL 700G NOTE 6

87 49R NOTE 6

NOTE 3

2000-F16-SCM-EL-10125 A13

A14 2000-F16-SCM-EL-10127

11kV 1250A 3-E 25kA FOR 3s

A15 A16 A17 A18 A19 A20

2000-F16-SCM-EL-10137

NOTE 2 NOTE 6A 1250A

2A 630A

VT
11kV/√3 0.11kV/√3 0.11kV/√3
15VA CLASS 0.2
25VA CLASS 6P

160E CLOSE

50 50G 81 AFD 51 51G PR SEL 751

50/1A 1.5VA 10P10

VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT

F24PW417-VS
HIGH PRESSURE PROCESS WATER PUMP 1
11/3.3kV 800kW VSD

F24PW418-VS
HIGH PRESSURE PROCESS WATER PUMP 2
11/3.3kV 800kW VSD

F17PP455-VS
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1
11/3.3kV 800kW VSD

F17PP456-VS
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2
11/3.3kV 800kW VSD

M 3~ RTD
800 kW

F24-PW-417
HIGH PRESSURE PROCESS WATER PUMP 1
2000-F16-SCM-EL-10131

F24-PW-418
HIGH PRESSURE PROCESS WATER PUMP 2
2000-F16-SCM-EL-10133

F17-PP-455
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1
2000-F16-SCM-EL-10135

F17-PP-456
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2
2000-F16-SCM-EL-10137

F16-TR-008
FLOTATION EMERGENCY 11/0.433kV TRANSFORMER

F16-SW-008
THICKENER EMERGENCY POWER 415V MCC INCOMER

ACTIVATION OF SWRM E-SHUT
DETECTION OF EARTH FAULT
DETECTION OF ARC FAULT

REFER DRG 2000-F16-SLD-EL-10221

↑CONT'D FROM SHEET 2 DRG 2000-F16-SLD-EL-10002

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 23/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER.
6. NORMAL OPERATING CONDITIONS :
   A) DCS MONITORS THE MAIN BUS VOLTAGE: UPON DETECTION OF MAINS LOSS, DCS TO INITIAL TRANSFER SEQUENCE.
   B) DCS TO OPEN BUSTIE CB, START GENERATOR AND CLOSE GENERATOR INCOMER CB WHICH ENERGISES THE RHS 11kV BUS.
   C) DCS MONITORS THE EMERGENCY BUS VOLTAGE TO CONFIRM POWER IS AVAILABLE.

   REINSTATEMENT - MAINS POWER RESTORED :
   A) DCS MONITORS THE MAIN BUS VOLTAGE: UPON DETECTION OF MAINS IS RESTORED, DCS TO PROMPT THE OPERATOR.
   B) OPERATOR TO INITATE TRANSFER SEQUENCE ON DCS.
   C) DCS TO STOP GENERATOR, OPEN GENERATOR INCOMER CB, CLOSE BUSTIE CB WHICH ENERGISES THE RHS 11kV BUS.
   D) DCS MONITORS THE EMERGENCY BUS VOLTAGE TO CONFIRM POWER IS AVAILBLE.
7. GENERATOR ENGINE START AND CONTROL CIRCUIT SHALL BE KEY INTERLOCKED WITH EMERGENCY BUS INCOMER CIRCUIT BREAKER EARTH SWITCH TO PREVENT STARTING OF GENERATOR WHILST EMERGENCY BUS INCOMER CIRCUIT BREAKER EARTH SWITCH IS CLOSED.
8. ONE SET OF PROTECTION CT's SHALL BE INSTALLED ON THE ALTERNATOR SIDE OF THE 11kV GENERATOR FOR WINDING PROTECTION.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
25	SYNCHRONISATION CHECK
27	UNDER VOLTAGE
46	REVERSE PHASE OR PHASE BALANCE CURRENT OR STATOR CURRENT UNBALANCE
49R	RTD BIASED THERMAL OVERLOAD
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
74 TCS	ALARM - TRIP CIRCUIT SUPERVISION
81	FREQUENCY
81O	OVER FREQUENCY
81U	UNDER FREQUENCY
86	LOCKING OUT
87	DIFFERENTIAL PROTECTION
AFD	ARC FLASH DETECTION

2000-F16-SLD-EL-10003_Rev1

Single line diagram sheet 3

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELVF16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SLD-EL-10003.DWG 09SEP25 6:51AM HOTSEAT04